

BURNETT COUNTY AP, WI, USA

WMO: 722183

Lat: 45.823N

Lon: 92.373W

Elev: 302

StdP: 97.76

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04972

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-23.9	-31.1	0.2	-26.0	-28.3	0.3	-22.9	9.3	-8.1	8.4	-8.4	1.3	310	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	30.9	21.7	29.1	20.4	27.7	19.4	23.3	28.4	22.2	27.0	21.2	25.7	4.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.1	26.3	20.7	15.9	25.3	19.7	15.0	24.2	70.8	28.1	66.8	27.0	62.7	25.6	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.6	6.9	DB	-31.2	33.9	2.4	1.8	-32.9	35.2	-34.3	36.3	-35.7	37.3	-37.4	38.7	(4)
(5)			WB	-31.4	25.2	2.4	1.2	-33.1	26.1	-34.5	26.8	-35.8	27.4	-37.5	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-10.2	-8.8	-1.0	5.9	12.8	17.9	20.8	19.0	15.3	7.9	0.5	-7.1	(6)	
	DBStd	12.14	7.40	6.95	7.05	5.39	4.98	3.72	3.45	3.61	4.73	4.99	6.08	6.70	(7)	
	HDD10.0	2598	626	526	347	143	26	1	0	1	9	101	287	532	(8)	
	HDD18.3	4661	884	759	600	372	183	51	16	34	114	324	533	790	(9)	
	CDD10.0	1201	0	0	6	22	114	238	333	280	168	37	4	0	(10)	
	CDD18.3	222	0	0	1	1	13	38	91	54	23	2	0	0	(11)	
	CDH23.3	2204	0	0	2	7	177	406	900	498	196	18	0	0	(12)	
	CDH26.7	559	0	0	0	0	49	91	267	111	40	1	0	0	(13)	
(14) Wind	WSAvg	2.8	2.8	2.9	2.9	3.2	3.1	2.6	2.5	2.3	2.7	2.9	2.9	2.7	(14)	
(15) Precipitation	PrecAvg	796	22	20	40	70	94	108	103	101	89	78	44	27	(15)	
	PrecMax	1067	55	45	80	203	170	153	166	238	191	176	129	52	(16)	
	PrecMin	570	4	2	17	15	29	49	28	38	21	23	2	7	(17)	
	PrecStd	124	13	12	17	39	39	33	38	46	48	44	35	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	10.0	21.3	24.4	31.4	31.9	34.8	31.4	30.5	26.0	18.8	7.9	(19)	
		MCWB	2.6	6.5	14.2	14.2	19.5	21.8	24.2	22.6	20.9	18.0	10.8	5.5	(20)	
	2%	DB	2.8	5.4	16.0	21.3	27.6	29.0	31.2	29.3	27.6	22.5	15.0	5.1	(21)	
		MCWB	1.1	2.4	10.8	11.6	17.6	19.3	22.4	20.9	19.5	15.5	9.9	3.2	(22)	
	5%	DB	1.4	2.7	12.2	18.2	25.1	27.4	29.5	27.6	25.9	18.9	12.3	2.8	(23)	
		MCWB	0.4	0.4	7.5	9.8	15.9	18.8	21.1	19.4	18.8	13.0	7.8	1.5	(24)	
	10%	DB	0.1	0.7	8.4	16.0	22.3	25.9	27.8	26.3	23.5	16.2	9.6	1.4	(25)	
		MCWB	-0.8	-0.8	4.8	8.5	14.4	17.6	20.1	19.1	17.7	11.0	6.3	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	6.3	15.6	15.2	22.4	23.6	25.7	24.1	23.2	19.9	13.2	6.7	(27)	
		MCDB	5.2	10.3	20.0	21.2	28.8	29.4	31.9	29.2	27.4	22.9	16.5	7.2	(28)	
	2%	WB	1.6	2.9	11.4	13.0	19.2	21.7	23.7	22.8	21.2	15.8	10.5	3.1	(29)	
		MCDB	2.9	5.5	14.8	18.7	24.7	26.9	28.9	27.0	25.4	20.7	14.6	4.5	(30)	
	5%	WB	0.4	0.8	7.9	11.1	17.5	20.3	22.5	21.6	20.0	13.7	8.1	1.7	(31)	
		MCDB	1.3	2.4	11.9	17.0	22.1	24.8	27.6	25.6	24.3	18.1	11.3	2.8	(32)	
	10%	WB	-0.8	-0.6	5.2	9.4	15.8	19.0	21.4	20.4	18.6	12.0	6.4	0.6	(33)	
		MCDB	0.1	0.5	8.2	15.2	20.9	23.7	26.2	24.6	22.7	15.1	9.8	1.3	(34)	
(35) Mean Daily Temperature Range	MDBR		9.6	10.6	10.9	11.8	13.0	12.4	12.0	11.9	11.7	10.2	8.8	8.3	(35)	
	5% DB	MCDBR	8.4	10.7	13.9	17.0	16.6	14.2	13.2	13.2	13.4	14.3	12.8	8.4	(36)	
		MCWBR	7.1	8.4	8.4	9.3	8.0	6.8	6.2	6.4	6.9	8.1	8.3	7.0	(37)	
	5% WB	MCDBR	8.1	9.4	12.2	14.0	13.1	11.7	11.4	10.4	11.4	11.5	11.4	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.252	0.268	0.313	0.359	0.392	0.383	0.377	0.389	0.363	0.315	0.286	0.263	(40)	
	taud		2.401	2.358	2.354	2.345	2.294	2.352	2.392	2.371	2.420	2.545	2.518	2.427	(41)	
	Ebn at Noon		873	922	917	899	875	884	884	859	853	859	826	817	(42)	
	Edh at Noon		70	89	105	117	127	121	115	113	99	75	64	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.65	2.75	3.86	4.63	5.33	5.87	6.21	5.18	3.98	2.48	1.55	1.15	(44)	
	RadStd		0.18	0.25	0.34	0.44	0.42	0.43	0.22	0.38	0.27	0.32	0.15	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page